SMARTDIET

Data Visualization of Physical Activity Patterns for Estimated Diet Needs

CDCS240 - Bachelor of Information Technology (Hons.)

Universiti Teknologi MARA
Fakulti Sains Komputer dan Matematik

INTRODUCTION

SmartDiet is an interactive data visualization dashboard that helps users understand how physical activity relates to estimated diet needs. The system integrates activity and nutrition datasets to calculate daily energy requirements based on user inputs such as age, gender, height, weight, and activity level. Through clear visuals and interactive features, SmartDiet supports exploration of calorie estimation, activity insights, and food comparisons for lifestyle awareness.

PROBLEM STATEMENT

Most fitness and diet tools provide fragmented information where activity tracking and diet planning are not integrated, making it difficult for users to balance calorie intake and expenditure. Even when users know how many calories they burn, they often receive no clear dietary guidance. Many applications operate independently and fail to combine activity and nutrition data in a meaningful way (Olyanasab & Annabestani, 2023). This issue is critical in Malaysia, where obesity and physical inactivity remain high (Institute for Public Health, 2023).

OBJECTIVE

1. To identify the data and system requirements for SmartDiet: Data Visualization of Physical Activity Patterns for Estimated Diet Needs.
2. To design SmartDiet: Data Visualization of Physical Activity Patterns for Estimated Diet Needs.
3. To develop SmartDiet: Data Visualization of Physical Activity Patterns for Estimated Diet Needs.

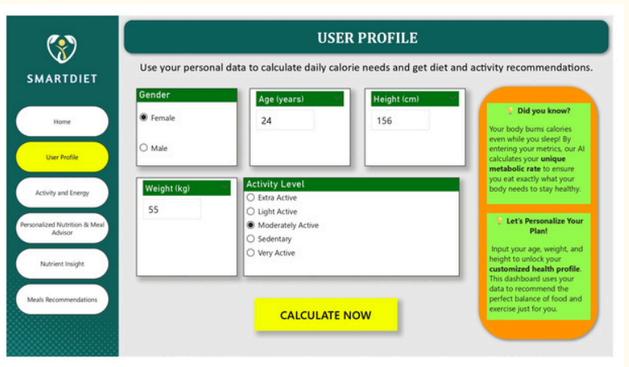
PROJECT AIM

The aim of this project is to develop SmartDiet, a user-friendly dashboard that supports personalized dietary planning based on physical activity patterns. The system integrates activity and nutrition datasets to estimate Total Daily Energy Expenditure (TDEE) and guide meal exploration aligned with individual calorie needs, helping users make informed lifestyle decisions.

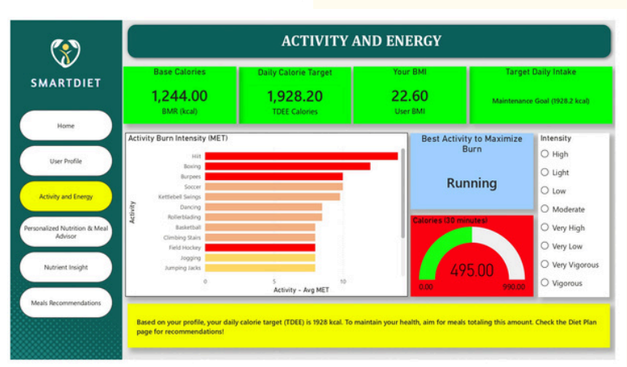
DATA SCOPE

- The Calories Burned dataset (2023) provides activity type, intensity level, MET values, and estimated calorie burn.
- The Food calorie dataset (2023) contains calorie values per 100g and serving size for various food items.
- The Fruits and vegetables nutrition dataset (2022) provides calorie and nutrient information.
- The Recipe dataset (2023) contains meal calorie and nutrition data for food exploration.

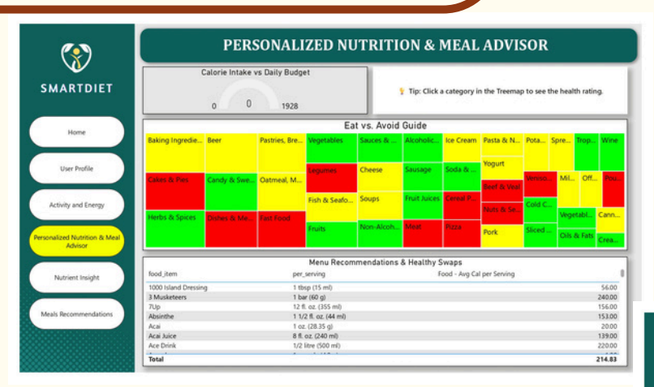
SYSTEM INTERFACE



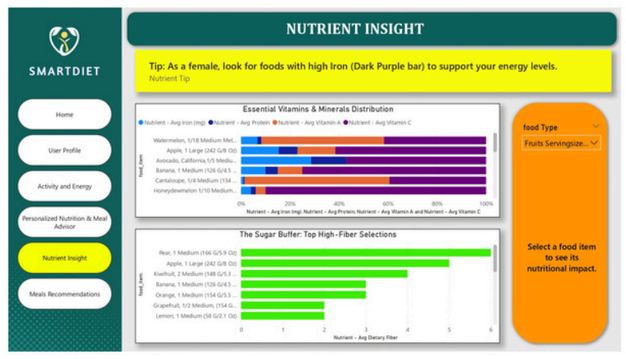
1) User Profile Page (Personal Data Input)



2) Activity and Energy Page



3) Personalized Nutrition & Meal Advisor Page



4) Nutrient Insight Page

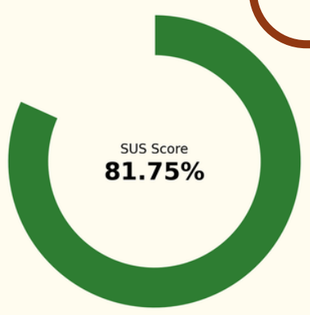
METHODOLOGY

The project applies a hybrid Agile–OSEMN methodology. Agile supports iterative dashboard development, while the OSEMN framework guides data processing through Obtain, Scrub, Explore, Model, and Interpret stages. Data cleaning and preparation were performed before integration into Power BI, where calculations and visual analytics were developed.

SYSTEM FEATURES

- Personalized calorie estimation
- Nutrient insights visualization
- Recipe and meal filtering
- Activity intensity comparison
- Interactive food calorie exploration
- User-driven dashboard controls

SUS TESTING RESULT



- 10 user feedback responses were collected to evaluate the SmartDiet system.
- The SUS testing produced an average score of 81.75.
- This score corresponds to a Grade A, indicating that the dashboard is rated as excellent, and the system is considered easy to use, efficient, and user-friendly.

CONCLUSION

SmartDiet successfully integrates physical activity and nutrition data into a single interactive dashboard. By visualizing energy balance, the system helps users make informed lifestyle decisions. Although not a medical tool, SmartDiet acts as an educational platform that promotes awareness of healthy habits and supports better daily choices.



PREPARED BY:
NURFARISHA BINTI MOHD KAMAL
2023103163
BIG DATA TRACK



SUPERVISED BY:
DR NOR DIANA BINTI AHMAD
nordiana@fskm.uitm.edu.my